

Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

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Abstract

Between 70–96 % of maritime casualties involve human error [1], and heavy weather makes coordination harder by degrading both perception and bridge-to-bridge radio. We present a tick-based agent-based model of Baltic and North Sea traffic. Vessels follow depth- and lane-constrained routes built from open bathymetry and Automatic Identification System (AIS) density data. Encounters are resolved with a Closest Point of Approach (CPA) check and a COLREGs-style give-way rule. The main modelling choice is that every collision warning succeeds only with probability equal to the product of a local weather factor and a crew-fatigue factor. Collision damage uses DTU SIMCOL [2]; route-network collision frequency is checked with IWRAP Mk II [3]; rescue follows IAMSAR practice [4] with MASSIM/Koopman random search [5, 6] and Ashrafi seasonal degradation [7]. We compare a classical baseline, a weather-aware baseline, and a fatigue-aware Proposed System across four scenarios (30 seeds each; 360 runs) with Mann–Whitney U tests, Holm–Bonferroni correction, and Cliff’s δ . Pre-registered fatal-rate and survival thresholds (H1–H3) were not confirmed after correction. Preparedness P_{prep} rose with large confirmed effects in every scenario ($|\delta| \geq 0.90$). Mean Storm Corridor collisions fell by about 28 % versus Baseline A ($\delta=0.32$, not significant after Holm).

Index Terms: maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

1. Introduction

Maritime transport moves most of world trade, but safety remains limited by human factors [1]. The European Maritime Safety Agency (EMSA) [8] reports weather as a frequent compounding factor in serious casualties, especially in congested coastal waters, and hull and machinery claims remain large [9]. The International Regulations for Preventing Collisions at Sea (COLREGs) [10] assume that crews see the encounter and can agree who gives way. That assumption breaks down under two effects that many maritime ABMs treat only loosely [11, 12, 13]:

- **Weather and radio.** Sea clutter and storms reduce the chance that a marine VHF warning is heard and acted on. Many models assume that once risk is detected, the manoeuvre always happens.
- **Crew fatigue.** Fatigue builds over a watch and follows a circadian pattern [14]. Prior work often keeps fatigue in a side model, without a direct link to whether a warning is executed.

This paper couples those effects in one product: warning success depends on both local weather and crew fatigue (Fig. 1).

Sections 3–6 define the Baltic–North Sea ABM, fatigue-aware CPA avoidance, SIMCOL damage [2], and SAR. Section 7 places network frequency on an IWRAP scale [3]; Sections 8–10 report the factorial comparison.

2. Objectives and Hypotheses

We test whether linking fatigue management to warning reliability reduces fatal outcomes more than classical, weather-unaware practice. We implement three configurations (storm-zone radio loss is shared by all methods—environmental, not a method privilege):

Baseline A (Classical) COLREGs give-way, no weather-aware slowdown, unmanaged fatigue.

Baseline B (Shore Rest Alerts) Weather-aware speed reduction and slower fatigue build-up; no further mitigation.

Proposed System Smart watch scheduling (lowest fatigue build-rate) on top of Baseline B.

Hypotheses:

- H1.** Proposed reduces the fatal-event rate (Table 3, KPI 1) by $\geq 20\%$ versus Baseline A ($p < 0.05$, Mann–Whitney U [15], Cliff’s $|\delta| \geq 0.2$ [16]).
- H2.** Proposed improves post-incident survival (KPI 4) by $\geq 15\%$ versus Baseline A.
- H3.** Under Storm Corridor, Proposed beats Baseline B on the fatal-event rate ($p < 0.05$).

3. Simulation Model

3.1. Domain and Time

The domain is the Baltic and North Sea: latitude $\phi \in [50.5^\circ, 66.0^\circ]$ N and longitude $\lambda \in [-5.0^\circ, 31.0^\circ]$ E. An equirectangular map with a mid-latitude correction sets one field unit to one nautical mile at $\phi_0 = 58.25^\circ$:

$$x = (\lambda - \lambda_{\min}) k_{\text{lon}}, \quad y = (\phi - \phi_{\min}) k_{\text{lat}},$$

with $k_{\text{lat}} = 60 \text{ nm}^\circ$ and $k_{\text{lon}} = 60 \cos \phi_0$. East–west distortion at the box edges is about 6–10 % and is left uncorrected; distances inside the engine are Euclidean nautical miles. Coastline polygons from Natural Earth sit in an R-tree for land and grounding tests. Bathymetry is used only offline for route planning (Section 3.3). Time steps are $\Delta t = 15 \text{ min}$ (0.25 h); the default run length is $T = 2880$ ticks (30 days). A wall-clock variable starts at 06:00 UTC and drives the circadian fatigue term (Section 4). Each run uses one seeded `SmallRng`, so the same seed yields identical KPI logs.

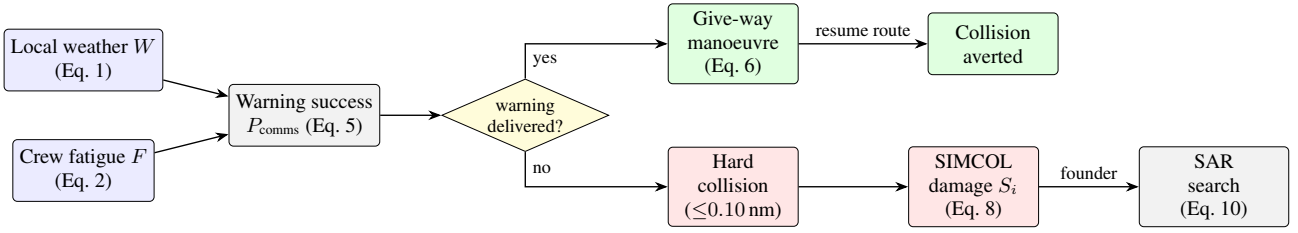


Figure 1: *Causal pipeline. Weather W and fatigue F set P_{comms} . A delivered warning yields a give-way manoeuvre; a missed warning leaves the encounter to geometry. Hard collisions feed SIMCOL; foundering enters SAR.*

3.2. Vessels and Voyage Logic

Agents are ships; after an incident, rescue assets appear (Section 6). Ship i has position $\mathbf{p}_i$, waypoint route R_i , travel direction $d_i \in \{+1, -1\}$, heading ψ_i , cruise speed v_i , state $\sigma_i \in \{\text{ACTIVE}, \text{DOCKED}\}$, crew $n_i = 10$, and fatigue $F_i \in [0, 1]$. Routes come from the offline network of Section 3.3 (data/ais_paths.json); ports are polygons in data/ports.json. Entering a port docks the ship. Cruise speed is drawn at spawn ($u \sim \mathcal{U}(0, 1)$):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / ro-ro / ro-pax} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases}$$

Waypoint pursuit. The active target is an avoidance waypoint if one is queued, otherwise the current route waypoint. A waypoint is reached within 0.5 nm. The step length is $s = \min(v_i \Delta t, \rho)$. Moves use up to five geometrically halved lengths so thin land is not tunnelled; a post-move check reverts grounding to the last valid position. At a port or route end the ship dwells $D \sim \mathcal{U}\{8, 48\}$ ticks, then reverses. Fleet size N is held fixed: when a ship is removed, a new one is spawned on a random synthesised track.

Same-destination following. Co-directional ships with end-points within 5 nm share a destination. A trailer within 3 nm of a leader matches the leader’s speed; within 0.5 nm it holds station. This avoids side-by-side overlap that the give-way rule does not separate.

3.3. Route Generation

Traffic geometry is produced offline so experiments stay reproducible. Depth comes from the EMODnet DTM [17]. Lane attractiveness comes from EMODnet vessel-density maps [18, 12]. Both are fetched once via the OGC Web Coverage Service (WCS) and stored as grids. Ports are a gazetteer of 42 harbours snapped to navigable water. Each class c has draft T_c and under-keel clearance (UKC $_c$) [19]. A cell of depth h is navigable iff $h \geq T_c + \text{UKC}_c + m_s$ with buffer $m_s = 2$ m (Table 1). Deep-draft very large crude carriers (VLCCs) are excluded (Danish Straits).

Between sampled port pairs, a weighted A* planner [20] runs on a 1 nm grid. Cell cost uses

$$\mu = 1 + w_h s_h + w_d s_d + w_\ell (1 - a_\ell),$$

penalising shallow margins (s_h), proximity to shore/shoal (s_d), and off-lane cells (a_ℓ). Default weights are $(w_h, w_d, w_\ell) = (3, 4, 2.5)$. Paths are simplified under a depth constraint and checked against raw bathymetry every 0.15 nm. Seeded cost

Table 1: *Minimum transit depth by vessel class (draft + under-keel clearance).*

Class	Draft T_c (m)	UKC $_c$ (m)	Required h (m)
Passenger	6.5	1.5	8
Cargo	11.0	2.0	13
Tanker	14.0	3.0	17

jitter keeps outputs deterministic per seed. The default network has 80 routes (35 cargo, 25 passenger, 20 tanker).

3.4. Weather Field

A 50×50 grid holds sea state s , visibility v , and wind u in $[0, 1]$. They fuse into hazard $W_{ij} \in [0, 1]$:

$$W = \text{clip} \left(\frac{w_s s + w_v (1 - v) + w_u u}{w_s + w_v + w_u}, 0, 1 \right), \quad (1)$$

with $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$. Each ship samples the nearest cell. The field advances each tick through a calm/storm regime chain, Lagrangian storm cells, AR(1) background fields, overlays, coupling, and smoothing (parameters in Appendix A). Independently, Storm Corridor places a moving storm of radius r_{storm} that drifts at 0.30 nm/tick along a fixed track (English Channel $\rightarrow$ North Sea $\rightarrow$ Skagerrak $\rightarrow$ Kattegat $\rightarrow$ Baltic $\rightarrow$ Gdańsk). Inside the zone, link reliability and non-A speeds are degraded (Sections 4 and 9.2; Table 2).

4. Crew Fatigue and Collision Avoidance

4.1. Fatigue Dynamics

Fatigue $F_i \in [0, 1]$ updates every tick. The circadian drive peaks near 03:00:

$$c(t_{\text{utc}}) = \frac{1}{2} \left(1 - \cos \left(\frac{2\pi(t_{\text{utc}} - 3)}{24} \right) \right) \in [0, 1].$$

While ACTIVE, with local hazard W ,

$$\Delta F = (\beta_{\text{base}} + \beta_W W + \beta_c c(t_{\text{utc}})) m(\text{method}), \quad (2)$$

with $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015, 0.0040, 0.0008)$ and

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A} \\ 0.82 & \text{Baseline B} \\ 0.65 & \text{Proposed,} \end{cases} \quad (3)$$

then $F_i \leftarrow \min(1, F_i + \Delta F)$. While DOCKED, recovery is 0.025 per tick. Above threshold $F^* = 0.40$, fatigue reduces

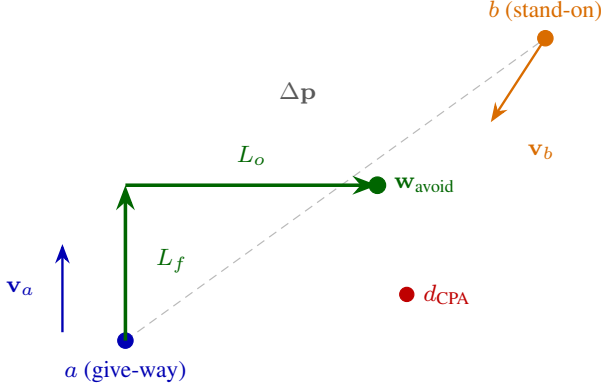


Figure 2: Give-way geometry. Ship *a* steers to a temporary waypoint $\mathbf{w}_{\text{avoid}}$ placed L_f ahead and L_o to starboard (Eq. 6).

warning reliability with maximum penalty $\kappa = 0.65$:

$$g(F_i) = \begin{cases} 1, & F_i \leq F^* \\ 1 - \kappa \frac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35, 1]. \quad (4)$$

4.2. CPA Detection and Give-Way

Each ACTIVE pair is screened with linear CPA. With relative position $\Delta \mathbf{p}$ and relative velocity $\Delta \mathbf{v}$,

$$t^* = -\frac{\Delta \mathbf{p} \cdot \Delta \mathbf{v}}{\|\Delta \mathbf{v}\|^2}, \quad d_{\text{CPA}} = \|\Delta \mathbf{p} + t^* \Delta \mathbf{v}\|,$$

where t^* is in ticks. A manoeuvre is considered only if $\|\Delta \mathbf{p}\| \leq 20$ nm, $d_{\text{CPA}} \leq 5$ nm, $0 < t^* \leq 12$ ticks, and the give-way ship is not in cooldown. The lower-id ship gives way (turn to starboard); the other stands on. This is a reproducible surrogate for COLREGs stand-on/give-way assignment [10, 11]. The manoeuvre runs only if the warning is delivered:

$$P_{\text{comms}} = \underbrace{\min(q(\mathbf{p}_a), q(\mathbf{p}_b))}_{\text{weather / storm}} \cdot \underbrace{\min(g(F_a), g(F_b))}_{\text{fatigue (Eq. 4)}}, \quad (5)$$

where $q(\mathbf{p})$ is nominal link reliability, reduced to the storm-comms rate when $\mathbf{p}$ is in the storm zone (0.08 in Storm Corridor). A uniform draw below P_{comms} executes the manoeuvre; otherwise geometry alone decides the encounter. A give-way ship receives one temporary waypoint $L_f = 3$ nm ahead and $L_o = 5$ nm to starboard of heading ψ (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f (\sin \psi, \cos \psi) + L_o (\cos \psi, -\sin \psi). \quad (6)$$

A short message log records the warning exchange; a 3-tick cooldown prevents thrashing.

4.3. Optional Force-Field Track

As an optional, default-off switch, the discrete waypoint of Eq. 6 can be replaced by a continuous repulsion field in the style of Xiao et al. [21]:

$$\mathbf{F}_{\text{rep}} = \sum_{k: d_k < d_o} \frac{k_{\text{rep}}}{d_k^p} \hat{\mathbf{n}}_k,$$

with exponent $p = 2$ by default. Source effect distances are micro-scale relative to a 15-minute basin tick, so magnitudes are rescaled while keeping the reported head-on to overtaking ratio. All definitive experiments keep this track off.

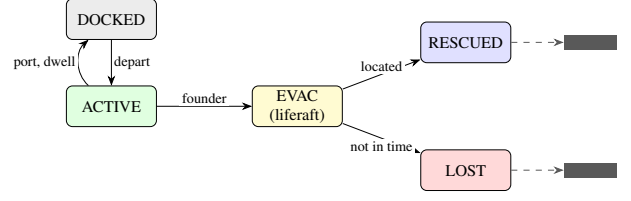


Figure 3: Vessel state machine. Foundering collision $\rightarrow$ EVAC; SAR yields RESCUED (Eq. 10) or LOST. Terminal ships respawn to keep fleet size.

5. Collision Consequence (SIMCOL)

A **hard collision** is logged when two ACTIVE ships close within 0.10 nm (≈ 185 m), once per encounter (rising edge). Contacts within 1 nm of a port are ignored as harbour manoeuvring under vessel traffic services (VTS). Damage uses DTU SIMCOL (ship collision damage / survivability) [2]. Absorbed energy follows Pedersen and Zhang [22, 2] on the normal closing component:

$$E_{\text{abs}} = \frac{1}{2} \frac{M'_a M'_b}{M'_a + M'_b} V_n^2, \quad V_n = |(\mathbf{v}_b - \mathbf{v}_a) \cdot \hat{\mathbf{n}}|, \quad (7)$$

with added-mass factors $m_{\text{sway}} = 0.85$ and $m_{\text{surge}} = 0.05$ [2]. Minorsky's correlation [23] links energy to damaged volume. Relative penetration t/B drives a survival factor shaped like SOLAS Chapter II-1 probabilistic damage stability [24]:

$$S_i = \begin{cases} 1, & t/B \leq r_0 \\ \left(\frac{r_1 - t/B}{r_1 - r_0} \right)^{0.25}, & r_0 < t/B < r_1 \\ 0, & t/B \geq r_1, \end{cases} \quad (8)$$

with $r_0 = 0.10$, $r_1 = 0.30$. This is an accepted surrogate: the engine does not carry per-ship residual-stability curves. Expected overboard count is

$$\mathbb{E}[\text{overboard}] = n_b (1 - S_i), \quad (9)$$

drawn as independent Bernoulli trials. If $S_i < S^\dagger = 0.50$, the struck ship founders and the whole crew enters the liferaft (EVAC, Section 6). If it stays afloat, only the overboard draw enters the man-overboard chain (Section 6.3). Encounter type is diagnostic, from the folded heading angle θ :

$$\text{type} = \begin{cases} \text{side-to-side (overtaking)}, & \theta < 45^\circ \\ \text{front-to-side (crossing)}, & 45^\circ \leq \theta < 135^\circ \\ \text{head-on}, & \theta \geq 135^\circ. \end{cases}$$

Near-parallel hits release little energy in Eq. 7; head-on and crossing hits drive deeper penetration.

6. Search and Rescue

Foundering starts the liferaft chain under International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual practice [4]. A rescue asset leaves the nearest port after a mobilisation delay of 2 ticks by default: helicopter (80 kn) beyond 40 nm offshore, otherwise patrol (25 kn). Persons overboard from afloat ships use a separate chain (Section 6.3). Figure 3 shows the vessel state machine. Grounding only reverts position; it does not force evacuation.

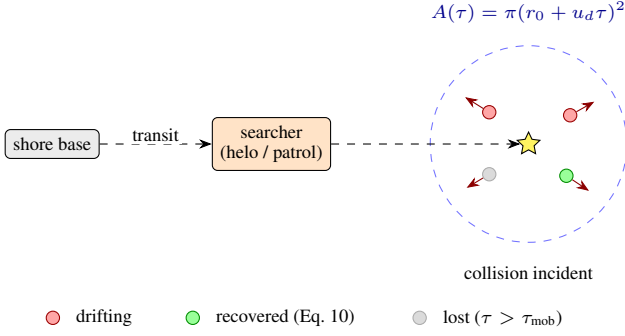


Figure 4: Man-overboard search. Casualties (Eq. 9) drift separately; one searcher applies Eq. 10 per person.

6.1. Drift and Detection (MASSIM)

Detection follows the Koopman random-search model used in the MASSIM maritime-search simulation [5, 6]. With sweep width w , search speed v_s , and area A ,

$$\text{CDP} = 1 - \exp\left(-\frac{n w v_s \tau}{A}\right) = 1 - e^{-C}. \quad (10)$$

Search area grows as $A(\tau) = \pi(r_0 + u_d \tau)^2$ with initial radius r_0 and drift u_d . Each on-scene tick draws Bernoulli detection at coverage $1 - e^{-dC}$.

6.2. Seasonal Degradation (Ashrafi)

Asset performance follows Ashrafi et al. [7]. Wind-chill uses

$$\text{wci} = 13.12 + 0.6215 T - 11.37 U^{0.16} + 0.3965 T U^{0.16}.$$

Months map to severity groups A–D (May–August mildest; Dec–Feb harshest). Relative transit/search speeds are approximately 1.00/0.95/0.77/0.60 (helicopter) and 1.00/0.80/0.40/0.21 (patrol); boarding times lengthen with severity [7]. Default simulation month is July (group A).

6.3. Man-Overboard Search

Overboard casualties (Eq. 9) form one incident of persons-in-water, each with its own drift bearing (Fig. 4), following Karatas et al. [5]. One searcher homes on the cluster centroid and applies Eq. 10 independently per person. Anyone not recovered within $\tau_{\text{mob}} = 12$ ticks (≈ 3 h) is lost. Liferaft endurance is 96 ticks (24 h).

7. External Collision-Frequency Check (IWRAP)

Internal KPIs compare methods. Network collision frequency is checked with IALA’s IWRAP Mk II [3, 25]:

$$N_c = N_a P_c.$$

Head-on, overtaking, and crossing candidate counts follow the usual IWRAP forms (flows Q , beams B , speeds V , Peder-sen diameter for crossings [3, 26]). Causation probabilities are the IWRAP defaults: $P_c^{\text{head-on}} = 0.50 \times 10^{-4}$, $P_c^{\text{overtaking}} = 1.10 \times 10^{-4}$, $P_c^{\text{crossing}} = 1.30 \times 10^{-4}$. Each batch row stamps N_c from the static synthesised routes, the scenario fleet size, and a method speed scale (weather-aware methods slow traffic). We do not export per-run AIS tracks. For the present network, N_c is on the order of 0.01 collisions/yr, inside the usual <1 /yr/fairway acceptance band.

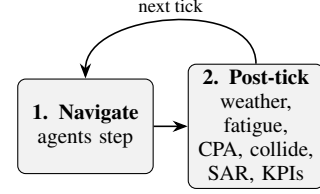


Figure 5: Macro-tick order. Agents navigate, then the post-tick pipeline advances weather, fatigue, CPA/comms, collisions, SAR/MOB, fleet top-up, and KPIs.

8. Experimental Design

8.1. Methods and Scenarios

The three methods of Section 2 differ only by fatigue build-rate m (Eq. 3), weather/storm speed awareness (Section 9.2), and preparedness weight α_M below. SIMCOL parameters are shared. Table 2 lists the four scenarios. Each condition uses 30 seeds and 2880 ticks (360 runs).

8.2. Key Performance Indicators

Table 3 defines the six hypothesis KPIs. Ship-hours add 0.25 h per ACTIVE ship per tick. A KPI *fatality* is an unrecovered person-in-water within τ_{mob} (Section 6.3); liferaft LOST is logged separately and does not enter KPI 1. Crew preparedness uses method awareness $\alpha_M \in \{2.0 \text{ (A)}, 1.3 \text{ (B)}, 1.0 \text{ (Proposed)}\}$:

$$P_{\text{prep}} = \max(0, 1 - \alpha_M W) \cdot \max(0, 1 - 0.70 F).$$

Diagnostic counters (collision-type mix, MOB tallies) are logged but not entered into the Holm battery.

8.3. Statistical Analysis

For each (KPI $\times$ scenario $\times$ method pair) we use a two-tailed Mann–Whitney U test [15] with Cliff’s δ [16]. Multiplicity uses Holm’s sequentially rejective Bonferroni procedure [27] over the full battery. A result is *significant* if $p_{\text{corr}} < 0.05$ and *confirmed* if also $|\delta| \geq 0.2$. Pairs are (A vs Proposed), (B vs Proposed), and (A vs B). Seeds are matched across conditions.

9. Implementation

The engine is Rust on krABMaga [28]. An Axum HTTP API launches factorial batches (Rayon) and streams progress with Server-Sent Events (SSE). Each finished run writes a JSONL tick log (every ten ticks), a manifest, and a SQLite results row. A React workbench (deck.gl / MapLibre) replays completed runs.

9.1. Tick Ordering

Within a tick, vessel agents step first (krABMaga order 200), then a post-tick agent (order 1000) runs the shared physics pipeline in Fig. 5; navigation therefore uses weather and avoidance state from the previous tick’s post-tick pass.

9.2. Speed Reduction

Baseline B and Proposed interpolate the just-moved position toward the last valid position by

$$f_W = \max(0.30, 1 - 0.45 W).$$

Table 2: *Scenario summary (30 seeds, 2880 ticks/run).*

Scenario	Key conditions	Ships	Hyp.
Calm Passage	Calm weather; calibration reference (target band ≈ 0.5 – 1.5 collisions / 1 k ship-hrs under Baseline A).	20	Calib.
Storm Corridor	Moving storm; storm-comms rate 0.08; non-A speed factor 0.45.	25	H1, H3
Blind Shore	Fleet-wide comms 0.55; tighter avoidance berth ($L_o = 3$ nm).	25	H1
Deep Water Rescue	Sparse fleet; SAR mobilisation 4 ticks; helo/patrol 55/16 kn.	15	H2

Table 3: *KPI definitions and hypothesis links.*

#	KPI	Definition	Hyp.
1	<code>fatal_per_1k_hrs</code>	Fatalities $\div$ ship-hrs $\times 10^3$	H1, H3
2	<code>collision_per_1k_hrs</code>	Collisions $\div$ ship-hrs $\times 10^3$	Calib.
3	<code>evac_activation_rate</code>	Evac activations $\div$ ship-hrs $\times 10^3$	—
4	<code>survival_ratio</code>	1 – fatalities / crew exposed in collisions	H2
5	<code>avg_tta_hours</code>	Mean rescue time-to-arrival	H2
6	<code>mean_p_prep</code>	Time-averaged crew preparedness	mech.

Table 4: *Pre-registered hypothesis outcomes (definitive 30-seed batch).*

Hyp.	Contrast	Verdict	Cliff’s δ
H1	Fatal rate, Proposed vs A	Not confirmed	$ \delta \leq 0.07$
H2	Survival ratio, Proposed vs A	Not confirmed	$ \delta \leq 0.07$
H3	Fatals under Storm, Proposed vs B	Not confirmed	$\delta \approx 0.00$

Baseline A ignores this factor. Inside the storm zone an extra factor applies: 1.0 for Baseline A, 0.45 in Storm Corridor for the other methods (default non-corridor storm factor 0.70 when a storm is enabled).

10. Results

10.1. Calibration and Storm Contrast

Confirmation criteria are those of Section 8.3. Figure 6 shows per-run KPI distributions; Figure 7 summarises effect sizes for the pre-registered contrasts. Under Baseline A, Calm Passage mean collisions were 0.60 per 1 000 ship-hours (inside the 0.5–1.5 band). Storm Corridor raised that rate to 1.48 ($2.48\times$ calm), restoring a clear weather contrast after the depth+lane route rebase.

10.2. Hypotheses H1–H3

Table 4 summarises the pre-registered contrasts. **H1** and **H2** are not confirmed: fatal rates and survival ratios sit near floor/ceiling in Calm Passage, Storm Corridor, and Blind Shore, so tests have little power. Deep Water Rescue has a higher fatal base rate; Proposed reduces mean fatalities from 0.038 (Baseline A) to 0.023 ($\approx 39\%$), but survival is almost unchanged and neither contrast survives Holm ($p_{\text{corr}} = 1$, $|\delta| \leq 0.02$). **H3** is non-significant (means 0.0022 vs 0.0024).

10.3. Mechanism and Collisions

Mean P_{prep} is higher for Proposed than for either baseline in every scenario, with confirmed effects ($|\delta| \in [0.90, 1.00]$); mean Storm Corridor collisions fall from 1.48 to 1.07 versus Baseline A ($\approx 28\%$, $\delta=0.32$), but that contrast is not significant after Holm.

11. Discussion

The core claim is Eq. 5: a warning succeeds only when weather and both crews allow it. The batch confirms the preparedness lever and a directional storm-collision benefit, but not the pre-registered fatal/survival thresholds. Fatal events are rare outside Deep Water Rescue, so H1–H3 are under-powered even when point estimates move in the expected direction. The survival factor S_i (Eq. 8) is a SOLAS-shaped surrogate, not a full residual-stability calculation. Main limitations: synthesised routes trade some traffic realism for control; the lower-id give-way rule simplifies COLREGs; force-field avoidance is off by default; IWRAP stamps use static routes rather than per-run tracks. Calm Passage calibration under Baseline A sits near published AIS near-miss bands [12, 13]. Longer runs or collision-rate primary endpoints would raise power on fatal contrasts.

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Definitive batch KPIs (30 seeds & 4 scenarios × 3 methods)

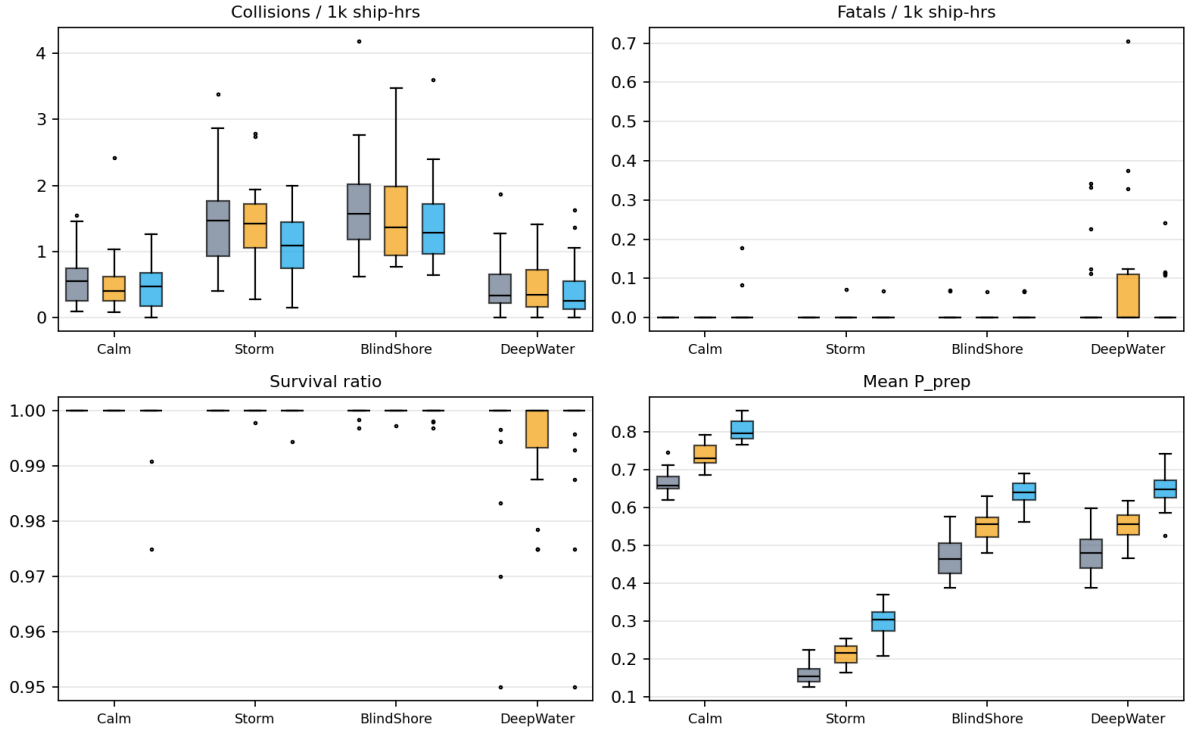


Figure 6: Per-run KPI distributions for the definitive batch (30 seeds). A = Baseline A, B = Baseline B, P = Proposed.

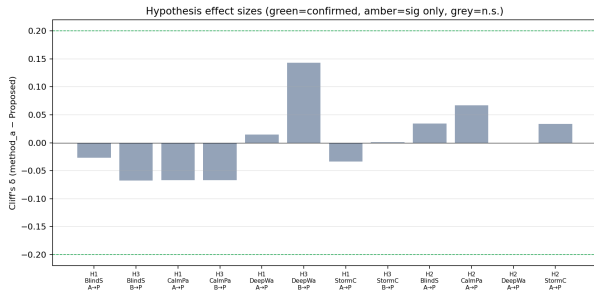


Figure 7: Cliff's δ for H1-H3-relevant pairs. Green: confirmed; amber: significant only; grey: not significant after Holm.

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A. Default Constants

Table 5 lists principal defaults (overridable in configuration).

Weather-field process (Mixed preset): regime flips $p_{cs} = 0.03$, $p_{sc} = 0.08$; Calm/Storm reversion targets $(0.30, 0.80, 0.25) / (0.72, 0.30, 0.75)$; up to 8 storm cells; AR(1) coefficients $(\rho_s, \rho_v, \rho_u) = (0.95, 0.90, 0.93)$, $(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$; gradient cap 0.12. Calm and Stormy presets scale the flip rates.

Table 5: *Selected engine default constants.*

Constant	Default	Role
Δt	0.25 h	tick length (15 min)
T	2880	ticks per run (30 d)
weather grid	50×50	hazard field
(w_s, w_v, w_u)	(1.0, 0.5, 0.8)	hazard fusion weights
warn / CPA / TTA gate	20 / 5 nm / 12 tk	avoidance gate
L_f, L_o	3 / 5 nm	give-way waypoint offsets
cooldown	3 tk	re-manoeuve lockout
r_{coll}	0.05 nm	hit at $2r = 0.10$ nm
port collision exclusion	1 nm	harbour water
follow vicinity / gap	3 / 0.5 nm	speed-match / keep-distance
same-destination	5 nm	endpoint match radius
$(m_{\text{sway}}, m_{\text{surge}})$	0.85, 0.05	SIMCOL added mass
Minorsky a, b	47.2, 32.7	energy-volume (MJ, m ³)
$\ell / L_{pp}, \tau$	0.12, 0.08 m	damage length / side steel
$r_0, r_1, S^\dagger$	0.10, 0.30, 0.50	S_i knee / loss / founder
$(\beta_{\text{base}}, \beta_W, \beta_c)$	0.0015, 0.0040, 0.0008	fatigue rates
recovery	0.025/tk	dock fatigue recovery
F^*, κ	0.40, 0.65	comms-degradation knee
storm-comms (corridor)	0.08	Storm Corridor link rate
storm-comms (generic default)	0.30	when storm enabled outside corridor presets
n crew	10	per vessel
dwell	8–48 tk	port stay
helicopter / patrol	80 / 25 kn	rescue asset speeds
Δt_{mob}	2 tk	rescue mobilisation delay
helo / patrol cutoff	40 nm	SAR asset-selection range
w, r_0, u_d	5 nm, 2 nm, 0.2 nm/tk	MASSIM sweep / datum / drift
survival window	96 tk	liferaft endurance (24 h)
τ_{mob}	12 tk	MOB cold-water window (3 h)
MOB scatter	0.5 nm	person-in-water spread
P_c (IWRAP)	$0.5\text{--}1.3 \times 10^{-4}$	causation probability